

The Standard Word Metric of B_3 Is Not Conditionally Negative Definite

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Result

Józiak asks whether the word metric of a general Artin group is conditionally negative definite (CND) [1, p. 2]. The following eight-point witness gives a negative answer to the universal reading of that question.

Theorem 1. *Let*

$$B_3 = \langle a, b \mid aba = bab \rangle$$

carry the word metric associated to $\{a^{\pm 1}, b^{\pm 1}\}$. This metric is not conditionally negative definite.

Here and below, capital letters denote inverses: $A = a^{-1}$ and $B = b^{-1}$.

Proof Intuition

The braid relation creates shortcuts among the second sphere of the Cayley graph. Split eight length-two braids into two four-point classes. Every pair inside either class is distance four, whereas sufficiently many cross-class pairs are only distance two. Giving the two classes opposite signs makes the within-class contribution dominate, producing a positive value of the Schoenberg quadratic form. The only bookkeeping issue is certifying the small word distances; an injective $\mathrm{SL}_2(\mathbb{Z})$ -plus-exponent coordinate makes that finite check exact.

Exact word-distance certificate

Define

$$\rho(a) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad \rho(b) = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix},$$

and let $e : B_3 \rightarrow \mathbb{Z}$ be exponent sum, so $e(a) = e(b) = 1$.

Lemma 1. *The homomorphism $\Phi = (\rho, e) : B_3 \rightarrow \mathrm{SL}_2(\mathbb{Z}) \times \mathbb{Z}$ is injective.*

Proof. The center of B_3 is generated by $z = (ab)^3 = (aba)^2$, and $B_3/\langle z \rangle \cong \mathrm{PSL}_2(\mathbb{Z})$. If $\Phi(g) = (I, 0)$, the projective image of g is trivial, hence $g = z^k$. Since $\rho(z) = -I$, the equality $\rho(g) = I$ makes k even; since $e(z) = 6$, the equality $e(g) = 0$ then forces $k = 0$. $\square$

Order the points as

$$(g_1, \dots, g_8) = (ab, BA, Ab, bA, ba, AB, Ba, aB).$$

Direct multiplication of their Φ -coordinates gives the following word-distance matrix:

$$D = (d(g_i, g_j))_{i,j=1}^8 = \begin{pmatrix} 0 & 4 & 4 & 4 & 2 & 4 & 2 & 2 \\ 4 & 0 & 4 & 4 & 4 & 2 & 2 & 2 \\ 4 & 4 & 0 & 4 & 2 & 2 & 4 & 4 \\ 4 & 4 & 4 & 0 & 2 & 2 & 4 & 4 \\ 2 & 4 & 2 & 2 & 0 & 4 & 4 & 4 \\ 4 & 2 & 2 & 2 & 4 & 0 & 4 & 4 \\ 2 & 2 & 4 & 4 & 4 & 4 & 0 & 4 \\ 2 & 2 & 4 & 4 & 4 & 4 & 4 & 0 \end{pmatrix}.$$

For completeness, this is a particularly small finite verification. Each point has word length two, so every displayed distance is at most four. Exponent parity makes every distance between these points even. Injectivity of Φ distinguishes the eight points; comparing $\Phi(g_i^{-1}g_j)$ with the twelve reduced words of length two decides whether the distance is two, and otherwise it is four. The included verifier performs exactly this enumeration using integer arithmetic.

Violation of conditional negative definiteness

Let

$$\lambda = (1, 1, 1, 1, -1, -1, -1, -1)^T.$$

Then $\sum_i \lambda_i = 0$, but the displayed matrix gives

$$\sum_{i,j=1}^8 \lambda_i \lambda_j d(g_i, g_j) = \lambda^T D \lambda = 8 > 0.$$

This contradicts the defining inequality for a CND kernel, completing the proof.

Scope and checks

The result refutes the assertion that standard word metrics of all Artin groups are CND; it does not classify which Artin groups have CND word metric. The source paper’s nearby “natural question” about quadratic embeddings is answered inside that paper by affine independence and is not the target here.

Cheap run indexes contained no prior packet for arXiv:1307.1778 or this Artin-word-metric question. Bounded exact-phrase searches through 11 August 2026 found the source question and literature on braid normal forms, but no later paper stating this eight-point obstruction or an answer for the standard B_3 word metric. Novelty therefore remains a candidate assessment, while the finite calculation itself is exact.

Reproduction

Run

```
conda run --no-capture-output -n sandbox python \
  code/verify_b3_witness.py
```

from this packet directory. The checked output is stored in `code/OUTPUT.txt`.

References

- [1] P. Józiak, *Conditionally strictly negative definite kernels*, Linear Multilinear Algebra **63** (2015), 2406–2418; arXiv:1307.1778, doi:10.1080/03081087.2015.1015401.